

Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

prof. Fabrizio Ferrandi

Advanced Algorithms and Parallel Programming – 2nd part (Parallel Programming)–June 30th 2022

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Questions 1 (4 points) _____

Questions 2 (4 points) _____

Questions 3 (4 points) _____

Questions 4 (4 points) _____

Questions n. 1

Answer the following questions about OpenMP and briefly explain (without explanation the answer will be considered invalid)

A. Describe the multithreading paradigm used in OpenMP and the lifespan of threads with respect to tasks.

B. How is variables visibility managed in OpenMP?

C. What is the "omp sections" directive used for?

D. Is it possible and/or mandatory to define barriers in OpenMP?

Questions n. 2

Answer the following questions about parallel algorithms and parallel programming and briefly explain (without explanation the answer will be considered invalid)

A. Describe the workpile pattern, situations where it may be applied and possible implementations.

B. Is the Quicksort algorithm easy to parallelize? Propose a high-level description of a possible parallelization and analyze its potential bottlenecks.

C. Describe how to implement a map pattern with one of the parallel programming models presented in the course.

Questions n. 3

Answer the following questions about MPI and briefly explain (without explanation the answer will be considered invalid)

A. Does MPI parallelize over thread or processes? What are the main differences between the two approaches?

B. Describe the behavior of the MPI_Reduce routine.

C. Are there any restrictions on the use of communication routines in MPI, particularly related to the use of other routines?

Questions n. 4

Answer the following questions about PThread and briefly explain (without explanation the answer will be considered invalid)

A. When a new thread is spawned and assigned a routine, is it possible to automatically pass some arguments to that routine?

B. What is the difference between joinable and detached threads?